GENERATIVE AI INFRASTRUCTURE · VIJAY SAI KILARI

What It Really Takes to Train a Large Language Model

03

The training process behind GPT, Llama, Gemini, Claude, and Grok, and what it costs in data, compute, energy, and time.

2.4×
yearly growth in frontier training costs since 2016

15.6T
tokens of text to pretrain Llama 3.1 405B

16,384
H100 GPUs used for that one training run

310 GWh
estimated to train Grok 4, about 28,700 US homes for a year

01 The training pipeline

Collect data 1
Web crawls, books, code, and licensed content, measured in trillions of tokens.

Clean and filter 2
Remove duplicates, spam, and personal data, then balance the mix of sources.

Tokenize 3
Split text into tokens, the word pieces the model reads and predicts.

Pretrain 4
Predict the next token trillions of times on thousands of GPUs. Most of the cost lives here.

Evaluate and checkpoint 5
Track loss, run benchmarks, and save progress through hardware failures.

Supervised fine tuning 6
Teach instruction following with curated, human written examples.

Align and red team 7
RLHF, Constitutional AI, or DPO, plus safety testing.

Deploy and serve 8
Every user request costs compute, long after training ends.

02 The four resources

Data

- GPT-3: 300B tokens (2020)
- Llama 3.1: 15.6T tokens (2024), about 50× more
- Llama 3 mix: 50% general, 25% math, 17% code, 8% multilingual
- News Corp content deal with OpenAI: reported $250M+ over 5 years

Compute

- Rule of thumb: 6 × parameters × tokens FLOPs
- GPT-3: 3.14 × 10^{23} FLOPs on ~10,000 V100s
- Llama 3.1 405B: 3.8 × 10^{25} FLOPs, 30.84M GPU hours
- Chips and staff: tens of millions $ each

Energy

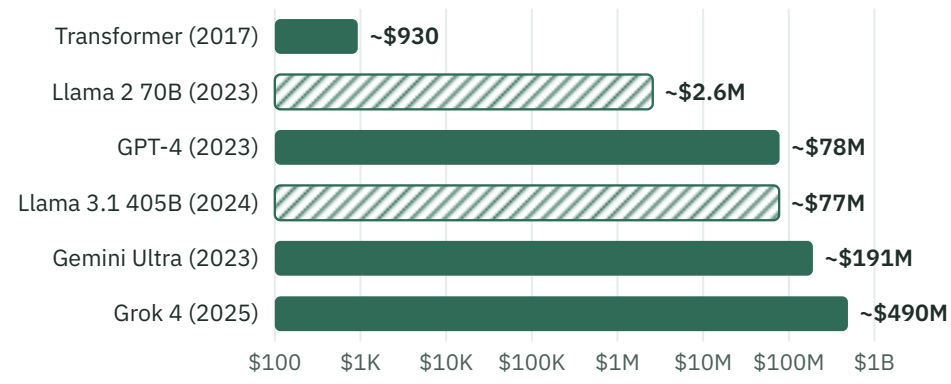
- GPT-3: 1,287 MWh, 552 t CO_2e
- Llama 3.1 405B: 8,930 t CO_2e (location based)
- Grok 4: ~310 GWh (estimate)
- Only 2 to 6% of the dollar cost

Time and people

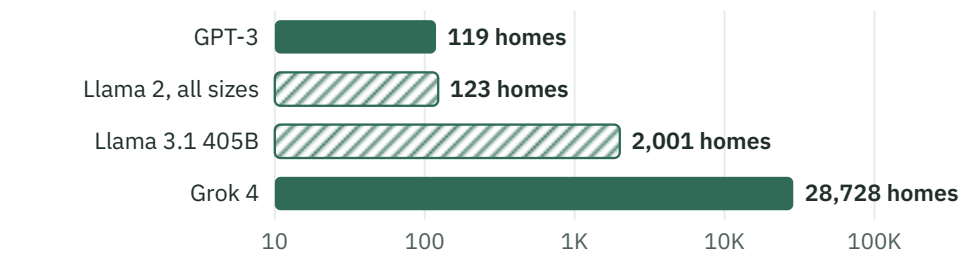
- GPT-3: 14.8 days of training
- Llama 3: 466 interruptions in 54 days, about one every 3 hours
- InstructGPT: ~40 contractors for human feedback

03 The numbers

Estimated training cost (log scale)
Hatched bars are my estimates from published GPU hours



Training electricity in US homes powered for a year (log scale)
Hatched bars are my estimates: GPU hours × GPU power



04 What each company has shared

MODEL	DATA	COMPUTE AND HARDWARE	ENERGY / EMISSIONS	COST	SOURCE TYPE
GPT-3 (OpenAI, 2020)	300B tokens	175B params; ~10,000 V100s; 14.8 days	1,287 MWh; 552 t	Not disclosed	Disclosed
GPT-4 (OpenAI, 2023)	Not disclosed	Not disclosed	Not disclosed	~$78M compute; CEO: over $100M	Estimate
Llama 2 70B (Meta, 2023)	2T tokens	1.72M A100 GPU hours	291 t (offset)	~$2.6M (my est.)	Disclosed
Llama 3.1 405B (Meta, 2024)	15.6T tokens	16,384 H100s; 30.84M GPU hours	8,930 t	~$77M (my est.)	Disclosed
Gemini 1.0 Ultra (Google, 2023)	Not disclosed	Large TPUv4 fleet, several data centers	Not disclosed	~$191M compute	Estimate
Claude 3.5 Sonnet (Anthropic, 2024)	Not disclosed	Not disclosed	Not disclosed	"A few $10M's"	Statement
Grok 4 (xAI, 2025)	Not disclosed	Grok 3: ~100,000 H100 Colossus cluster	~310 GWh; ~150,000 t	~$490M	Estimate

05 Takeaways

Pretraining dominates
Most chips, energy, and time go to one step.

Costs outrun budgets
Largest runs projected past $1B by 2027.

Energy is the constraint
Small share of dollars, big limit on where clusters go.

Transparency is uneven
Only Meta publishes detailed numbers.

Build on, do not train
Fine tuning costs a tiny fraction of pretraining.

Sources: Brown et al. (2020); Patterson et al. (2021); Ouyang et al. (2022); Bai et al. (2022); Meta Llama 2 and 3.1 model cards; Llama Team (2024); Gemini Team (2023); Stanford AI Index (2024); Cottier et al. (2024); Amodei (2025); xAI (2024); Tom's Hardware (2025); Epoch AI (2025); U.S. EIA. Full references on the interactive page.

Interactive version
omsairamkvs99.github.io/ml-portfolio/artifacts/03-llm-training-costs/